Pre-Match Batter Milestone Predictor

Session 02 | Development model and 2025 chronological validation | 20 October 2026

Scope: Estimate the chances that an IPL batter scores at least 30 or at least 50 runs, conditioned on having a recorded batting innings. This is not yet a prediction of playing-XI inclusion, batting appearance, or exact score.

17,316	1,085	35 / 35
Eligible historical innings	2025 validation innings	Automated tests passed

Summary of measured findings

2025 metric	Selected Random Forest	Smoothed player-form baseline
Brier - 30+ (lower better)	0.194459	0.197376
Brier - 50+ (lower better)	0.109510	0.111076
Mean of both Brier scores	0.151985	0.154226
Three-class log loss	0.772682	0.793569

Interpretation. The selected model modestly outperformed smoothed historical player-form frequencies on the 2025 validation season. The model was chosen using that same season, so this is development evidence, not an independent generalization test. IPL 2026 remains fully reserved for Session 03.

Source: Men's IPL ball-by-ball Cricsheet JSON archive processed in Stages 1-2. Super Over deliveries are excluded from regular batting records; no original data is overwritten.

01 / DATA AND PRE-MATCH FEATURES

The training unit is one recorded batter innings per player and match. We create 3 exclusive labels: fewer than 30 runs, 30-49 runs, and at least 50 runs. The model estimates their probabilities together.

Data screening item (years 2008-2025)	Count
Recorded batter-innings rows	17,638
Exclude D/L-marked match innings	263
Exclude additional no-result match innings	59
Eligible rows retained	17,316
Fit rows through 2024	16,231
2025 validation rows	1,085
Distinct eligible player IDs	702

How each feature is constructed

At a match date, look only at strictly earlier calendar dates for that batter. Calculate previous innings count, career runs and milestone rates, last 5 and last 10 performances, histories against the chosen opponent and at the ground, and time since last appearance. Team, opponent and announced venue are known before the match.

No future leakage: Today's batter runs, balls faced, dismissal, real batting position, toss outcome and innings score are not features. We calculate all examples for a day before adding that day's results to historical form. The 2026 season is excluded before any form calculations.

Exceptional matches and scope

Initial modeling excludes D/L-marked and no-result matches for a more consistent cohort; this may introduce sample-selection bias, so the choice is documented and revisitable. The ordinary innings of Super Over-decided ties remain eligible, but their Super Over deliveries never become standard batting innings. These exclusions do not remove games from historical EDA.

Important modeling assumption

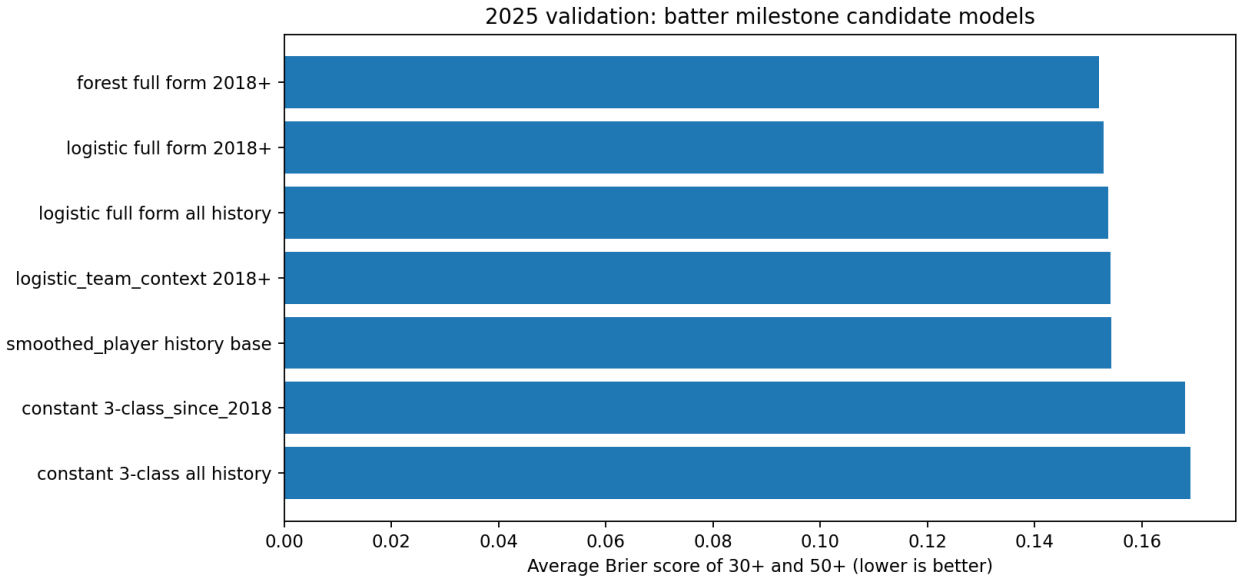
A row exists only when the batter is recorded as having batted. Therefore these probabilities are conditional on a batting innings. A player omitted from the playing XI or who does not bat must not be scored as if they failed to reach a milestone. A participation model would be a separate task.

02 / MODEL AND EVALUATION

Why one model for two milestones? If classes 0, 1 and 2 are <30, 30-49 and 50+ respectively, then P(30+) = P(class 1) + P(class 2), while P(50+) = P(class 2). Hence P(50+) can never be greater than P(30+).

Development design

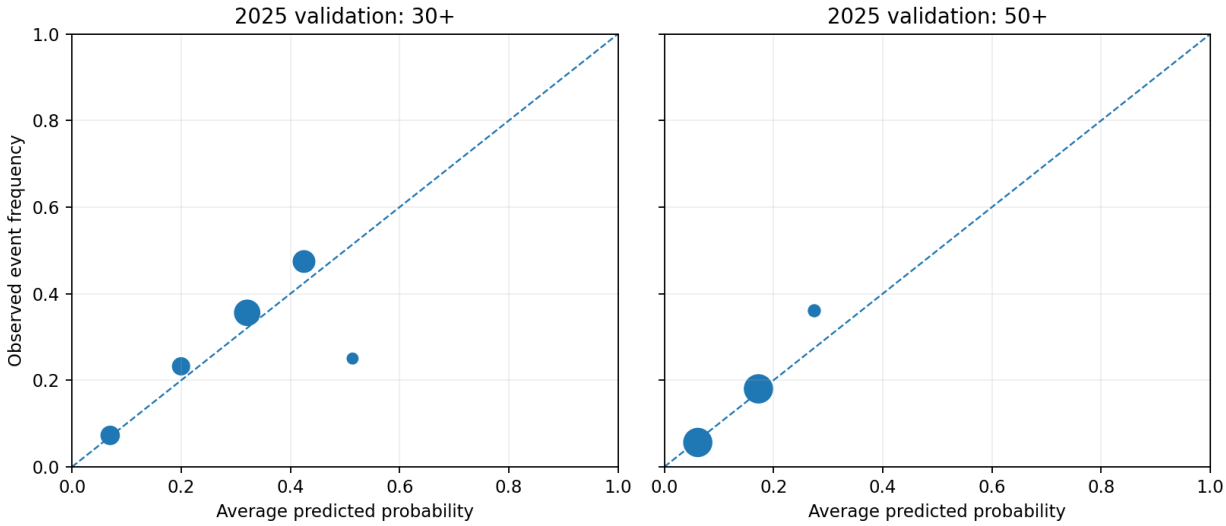
Fit candidate Logistic Regression and bounded Random Forest classifiers using match dates through 2024. Validate on 2025 with unchanged model parameters, allowing historical features for a later 2025 match to incorporate earlier already-completed 2025 innings. Compare constant class rates and player-form smoothed baselines. Select using average of the two milestone Brier scores.



Selected: Random Forest using labeled examples from 2018-2024 (6,918 training innings) with pre-match player history, venue and opposition. On 2025: ROC-AUC 0.6954 for 30+; 0.7319 for 50+; multiclass log loss 0.7727. AUC is supplementary and does not replace calibration.

03 / RELIABILITY AND LIMITATIONS

Calibration and sample support

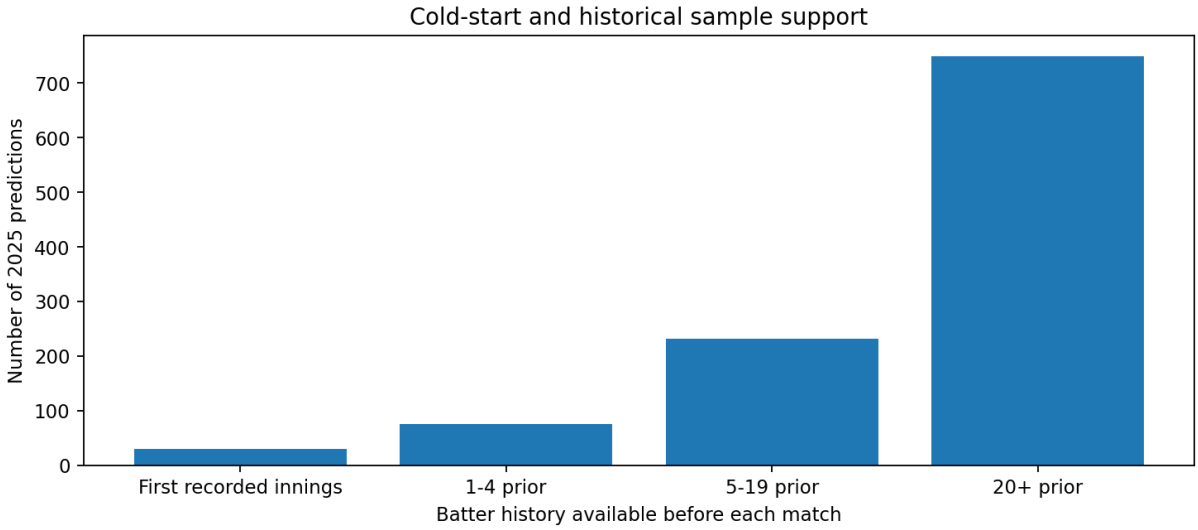


Calibration compares forecast probabilities with observed event frequencies in bins. Some bins are sparsely populated and should not be overinterpreted. The match-cluster bootstrap mean-Brier improvement over smoothed player form is +0.002241, with a descriptive 95% percentile interval [+0.000397, +0.003978]. Selection on this same 2025 season makes this an optimistic post-selection comparison, not a final guarantee.

Uncertainty the website must explain

Thirty 2025 validation innings were by batters with no earlier eligible IPL innings. Such cold-start forecasts require caution. The model does not know confirmed playing XI, availability, whether a player will bat, real in-match batting order, or weather changes. These data are not assumed to be available before the match.

04 / REPRODUCE, DEFEND, NEXT



How to reproduce locally

Step	Command or path
Build pre-match features	python -m cricketlens.batter_training_data
Fit models; validate 2025	python -m cricketlens.fit_batter_models
Generate plots and bootstrap	python -m cricketlens.stage03_batter_diagnostics
Run the complete test suite	pytest -q
Historical example (2025)	python -m cricketlens.predict_batter --match-id 1473438 --player-id 29e95537

What a viva examiner might ask

1. Why three classes? Coherent nested milestone probabilities. 2. How avoid leakage? Strictly earlier-calendar-day form only and whitelisted features. 3. Why not call a no-show a 0? The target is conditional on batting. 4. Why compare baselines? To justify value beyond simple historical rates. 5. What remains? Freeze the specification, evaluate once on 2026, and then refit through 2026 separately for future use.

Next: Stage 3 / Session 03

Confirm model definitions and baselines, evaluate both models on the unseen 2026 test set exactly once, report calibration and cold starts, and separately train the deployment versions on verified data through 2026. Do not merge Stage 3 into main until review and completion.

Repository documentation: docs/STAGE_03_SESSION_02_BATTER_MODEL.md and docs/STAGE_03_VIVA_QUESTIONS.md. The raw archive, generated training data and .joblib models remain local, ignored by Git.